AIMLCZG512 - Deep Reinforcement Learning (4 U)

Session-wise Coverage Notes (in brief) - Up to Mid-Term Exam

Text Book:

(1) Most part of this course refers to TB-1 - Reinforcement Learning – Introduction, Sutton & Barto (https://www.andrew.cmu.edu/course/10-703/textbook/BartoSutton.pdf)

(2) Certain topics refers to research publications. This is mentioned in the table below. The specifics on the coverage requirements from the research papers will be discussed in the course meetings

Course Meeting Schedule:

- Every Tuesday 8:00 PM to 9:00 PM
- Recurring meeting invite is shared with all the instructors

Session-wise plan:

Class #1 – Introduction to RL

- Begin with a short walk-through of the **course handout** and explain how this course is the *first-level* foundation for Deep RL.
- Cover **Chapter 1 (Sutton & Barto)** – define what is RL, explain the **four elements** (policy, reward, value, model). Keep the level light here; remind students that we'll revisit these ideas later in more depth.
- Spend about an hour on the **Tic-Tac-Toe example** from the textbook — explain how RL ideas emerge through this simple game. Encourage questions like "How is this different from supervised learning?" & use this example to make terms like *agent*, *environment*, and *reward* concrete.

Class #2 & #3 – Multi-Armed Bandits (MAB)

- Start with the *motivation*: why do we need the bandit setup? Bring real examples (e.g., ad recommendation, product trials).
- Explain the **MAB problem definition**, introduce **action-value estimation**, **incremental updates**, and **exploration vs exploitation trade-off**.
    - From the book (Chapter 2), discuss Sample average and constant step-size methods, Stationary **vs non-stationary** settings, Optimistic **initial values**, **ε-greedy**, and **UCB** methods.
- In Class #3, show the logic behind the **gradient bandit method** (just the idea, not full derivation).
- Wrap up with **Associative Search** & use examples from contextual bandits to make the concept relatable.

- **Note:** I recommend you to indeed plan MAB for two sessions, at a slower pace dealing concepts. This will feel long for sure. But it's important to be more detailed here. Students must have good conceptual grip before they move to MDP

Class #4-5 – Markov Decision Processes (MDP)

- Introduce the formal MDP framework (refer to **Chapter 3**.)
- Cover *states, actions, rewards, transitions*, and how they together form a **Markov process**.
- Spend good time on the **Bellman expectation & optimality equations**. Explain these slowly one term at a time. Use the PPT to guide your level of depth; avoid rushing through.
- Encourage drawing small state diagrams & help understand the backup diagrams. Explain how the Bellman equation connects current value to expected future return.

Class #5-6 – Dynamic Programming (DP)

- Move to **Chapter 4**: start from the idea that if we know the model, DP can compute the optimal policy.
- Discuss **Policy Evaluation, Policy Improvement, Policy Iteration** and **Value Iteration, Generalized Policy Iteration (GPI), Asynchronous DP**; Use the PPT + TB contents.
- The goal here is not to prove every little thing in this module but to make students understand how these methods iteratively improve value and policy.
- Reinforce the intuition of *bootstrapping*: updating estimates based on other estimates. This idea will be used throughout the course.

Class #7 – Monte Carlo Methods (On-Policy)

- (Refer **Chapter 5**) introduce Monte Carlo prediction and control. Explain the meaning of *first-visit* and *every-visit* MC. Explain how return averages lead to value estimates.
- Discuss the *Exploring Starts* assumption and transition to **ε-soft policies**.
- Introduce Off-policy Monte-Carlo

Class #8 – Monte Carlo(off policy) + Review of mid sem portions

- Finish remaining MC topics from the previous class Particularly on Off Policy Montecarlo (use the commonly agreed PPT as the guide to decide the depth)
- Start Temporal **Difference Learning (Chapter 6)**. Compare MC and TD approaches & focus on the idea of *bootstrapping vs complete return*. Cover **TD (0)** and **SARSA** in detail; use small examples from the book.
- Give a summary **Mid-Term Exam**

Guidelines for Instructors:

(1) All the lecture session will happen parallelly. It assumes that all the faculty delivering the common course material. Deviating from the shared material is not permitted.

Any additional topics, different approach that you intend to take must be discussed in the meeting.

(2) Meeting Attendance is mandatory. Following is the plan for PPT Review / Update / finalization:

Course Materials

- Refer to the existing PPT's and align with the Text book contents
- Provide all the necessary + References in the PPT
- For all the topics, Use the Sutton & Barto as the primary reference.

Quiz Question Contribution (3 questions per session): The quiz questions that you deliver (in the quiz question excel sheet) with utmost clarity & technical correctness. Topics must be within the scope of our coverage – Caution: These questions will be directly populated in the Quiz bank and be used for exam. Minor mistakes can lead to serious issues.

EC 2 & EC 3 Question Contribution (1 each every module) : The exam questions that you deliver (in the quiz question excel sheet) with utmost clarity & technical correctness. Topics must be within the scope of our coverage – Caution: These questions will be directly populated in the question bank and be used for exam. Minor mistakes can lead to serious issues. Please ensure all the details like values of parameters, initializations are given correctly.